Lecture 5.3: Managing Data for Reproducibility: Git I

AREC 570 / ECON 530 / Unit 5

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Reading and today's path

Gentzkow and Shapiro, *Code and Data for the Social Sciences*, Sections 1–4 (Gentzkow and Shapiro 2014)

Bryan, *Excuse Me, Do You Have a Moment to Talk About Version Control?*, 2017 (Bryan 2017)

1. Organize a project so another researcher can follow it
2. Identify where undocumented memory enters the workflow
3. Set up Git, GitHub, and RStudio
4. Take one research project from its first commit through GitHub
5. Choose a workflow for team projects and reusable templates

Reproducibility starts with organization

Can another researcher reconstruct a result from the same inputs?

They need

- Original inputs
- Code for every transformation
- The order in which the code runs
- Software and package requirements
- The project version used for the result

Source: Gentzkow and Shapiro (2014)

Start with a recognizable project root

```
research_project/
  README.md
  research_project.Rproj
  run_all.R
  code/
  data/
  output/
  docs/
  lab_diary.md
```

The `.Rproj` file opens this folder as the project root

Separate files by function

```
code/                   scripts and reusable functions
data/raw/               source files as retrieved
data/derived/           files constructed from source data
output/figures/         figures produced by code
output/tables/          tables produced by code
docs/                   codebook and project documentation
```

Raw inputs remain unchanged. Scripts rebuild derived data and outputs

Paths should travel with the project

Use paths relative to the project root

```
read.csv("data/raw/streamflow.csv")
```

A path beginning with one person's user directory usually fails on another computer

```
read.csv("C:/Users/jane/Desktop/streamflow.csv")
```

Automate the sequence

`run_all.R` records the order of operations

```r
source("code/00_setup.R")
source("code/01_get_cdl.R")
source("code/02_get_streamflow.R")
source("code/03_get_weather.R")
source("code/04_get_acs.R")
source("code/05_build_daily.R")
source("code/06_outputs.R")
```

One command rebuilds the project from documented inputs

Where does memory enter the workflow?

Consider one result in your current research project

- Which steps happen through code
- Which steps happen through menus or manual edits
- Where the order of operations is recorded
- Which inputs can change over time
- What another researcher would need to ask before running it

Documentation replaces memory

Different records answer different questions

Record	Question it answers
Scripts and `run_all.R`	What operations produced the result
README	How should a new researcher run the project
Codebook	What do the variables and values mean
Retrieval log	Where and when did the inputs originate
Lab diary	Why did the researcher make a decision
Version history	What changed across project states

What version control adds

Version control gives a project a documented history. At selected points, it records the state of the project's files together with the author, date, and reason for the change.

That history allows a researcher to compare versions, recover earlier work, and identify the exact project state used for an analysis or manuscript.

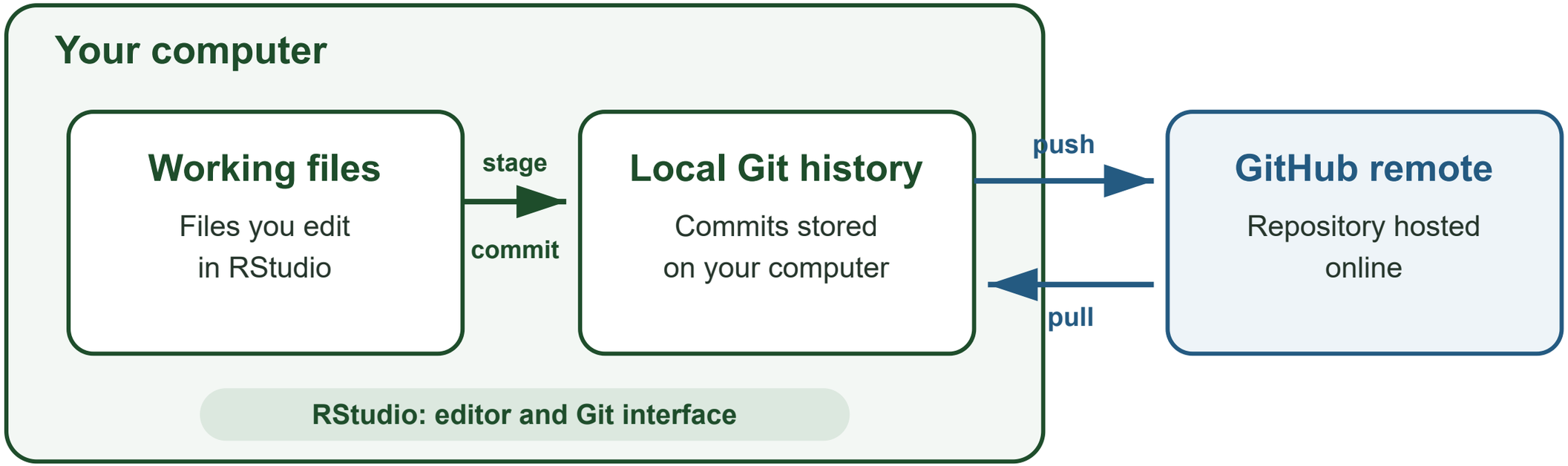
Source: Bryan (2017)

Git, GitHub, and RStudio have different roles

Tool	Role
Git	Software that records the history of a project on your computer
GitHub	An online service that hosts a copy of a Git project and supports sharing
RStudio	An editor that lets you work with project files and run Git through buttons, menus, and the Terminal

A Git project can remain entirely on one computer. GitHub adds an online copy. RStudio calls Git commands on your behalf (Bryan 2017)

One project can exist in three locations



Version control vocabulary

Term	Meaning
Repository	A project and its Git history
Working directory	The files currently visible in the project folder
Tracked file	A file whose changes Git follows
Staging area	The changes selected for the next commit
Commit	A recorded checkpoint in the project history
Diff	The lines added, removed, or changed

Version control vocabulary

Term	Meaning
Commit message	A short explanation attached to a commit
Tag	A readable name attached to an important commit
Local	The repository on your computer
Remote	A connected repository hosted online
Origin	The default name for the remote connected to the local repository
Push	Send local commits to a remote
Pull	Bring remote commits into the local repository

Saving, committing, and pushing are separate actions

Save writes the current file to the working directory

Commit records selected changes in the local Git history

Push sends local commits to GitHub

One-time setup

Complete these steps once on each computer

1. Install Git
2. Create a GitHub account
3. Tell Git your name and email
4. Create a GitHub personal access token
5. Store the token for RStudio and Git
6. Verify the connection

The demo repository provides detailed instructions in `docs/setup-rstudio.md`

Step 1: install Git

Download Git from git-scm.com/downloads

Restart RStudio after installation, then run

```
1  system("git --version")
```

A successful check prints a version such as `git version 2.51.0`

Steps 2 and 3: create accounts and identify yourself

Create an account at github.com

Install two helper packages and configure Git

```
install.packages(c("usethis", "gitcreds"))

usethis::use_git_config(
  user.name = "Your Name",
  user.email = "you@example.edu"
)
```

Use an email connected to your GitHub account

Steps 4 and 5: connect GitHub

Create a personal access token

```
usethis::create_github_token()
```

Copy the token from GitHub, then store it

```
gitcreds::gitcreds_set()
```

Treat the token like a password. Keep it out of scripts, `.Renviron`, and commits

Step 6: verify the setup

Run

```
usethis::git_sitrep()
```

Check for

- The expected Git version
- Your name and email
- A stored GitHub token
- A recognized GitHub account

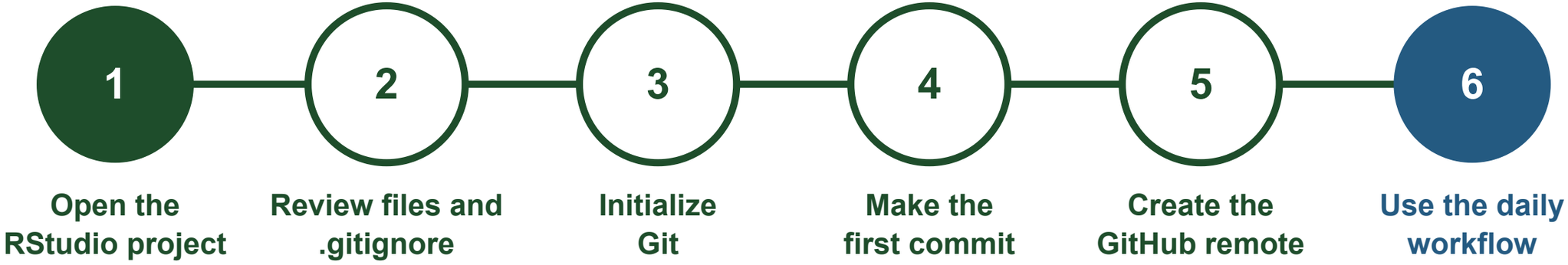
Case 1: your own research project

You have an existing RStudio project with scripts, data documentation, and early results

Your goal is to

1. Review which files belong in Git
2. Initialize the local repository
3. Make the first commit
4. Create a private GitHub repository
5. Use the workflow during each research session

Start-to-finish workflow



Open the project in RStudio

Open the `.Rproj` file inside the project root

Confirm that

- The project name appears in the upper-right corner
- The Files pane begins at the project root
- Relative paths run successfully
- The project contains a README or equivalent instructions

Decide what Git should track

Track files that explain, run, or report the research

- Source code
- README files and project documentation
- Data dictionaries and retrieval logs
- Small configuration files without credentials
- Manuscript and presentation source files

Generated outputs can follow the project team’s publication and archival policy

Use `.gitignore` for excluded files

Common exclusions include

- API keys and credentials
- Confidential or restricted data
- Large source files stored in an archive
- Derived files rebuilt by code
- R session files such as `.RData` and `.Rhistory`

`.gitignore` applies to files Git has not started tracking

Check before the first commit

Open `.gitignore` and review every file in the project

- Does any file contain a password or API key
- Does the project contain restricted or licensed data
- Is a large file available from a stable archive or reproducible download
- Can a derived file be rebuilt by the project scripts
- Does each tracked file help explain, run, or report the research

Revoke any credential that enters a commit because earlier commits retain its contents

Initialize the local repository

Run in the RStudio Console

```
1  usethis::use_git()
```

RStudio may restart the project. A **Git** tab should then appear beside Environment and History

This command creates the hidden `.git` directory that stores local history

Git records changes through three states

State	Meaning
Working directory	Files currently being edited
Staging area	Changes selected for the next commit
Commit	A recorded checkpoint with an explanatory message

Staging lets one work session become several coherent commits

Read the RStudio Git pane

Status	Meaning
`M`	A tracked file was modified
`?`	A new file is untracked
`A`	A new file is staged for its first commit
`D`	A tracked file was deleted
`R`	A tracked file was renamed or moved

Diff shows changed lines. The **Staged** boxes select changes for the next commit

Make the first commit

1. Open the Git pane
2. Select **Diff** and inspect the proposed files
3. Stage the files that belong in the repository
4. Enter `Create initial project structure`
5. Select **Commit**

The commit now exists in local Git history

Create the private GitHub repository

Run in the RStudio Console

```
usethis::use_github(private = TRUE)
```

This command

- Creates a GitHub repository under your account
- Connects the local repository to the GitHub remote
- Names that remote `origin`
- Pushes the current branch and commits

The everyday Git workflow

1. **Pull** before beginning a work session
2. Make one coherent change
3. Inspect the changed lines with **Diff**
4. Stage the files that belong to the change
5. Write a specific message and **Commit**
6. **Push** the commits to GitHub

Repeat the edit, inspect, stage, and commit steps as the work develops

Follow along in RStudio

In a practice project

1. Add one sentence to `README.md`
2. Save the file
3. Inspect the changed line with **Diff**
4. Stage `README.md`
5. Commit with `Clarify project research question`
6. Push the commit
7. Open GitHub and locate the same commit

The corresponding Git commands

```
git pull
git status
git diff
git add README.md
git diff --staged
git commit -m "Clarify project research question"
git push
```

The RStudio buttons run these commands

Companion reference: `docs/git-in-rstudio.md`

A diff shows what changed

```
- This project studies water use
+ This project studies how irrigation restrictions affect water use
```

The diff records the changed text

The commit message records the reason for that change

```
Clarify treatment and outcome in research question
```

Write messages for future readers

General message	Research-specific message
`updates`	`Code suppressed ACS estimates as missing`
`fixed bug`	`Keep FIPS codes as text to preserve leading zeros`
`final version`	`Restrict sample to the April–September growing season`

A useful message identifies the completed change. Add an explanation when a change alters the sample or interpretation

History makes earlier states inspectable

Use **History** in RStudio or **Commits** on GitHub to inspect

- The files changed in a commit
- The exact lines added and removed
- The author and date
- The commit message
- The identifier for the checkpoint

Tags such as `5.2-retrieve` give a readable name to a particular commit

Case 2: shared work and reusable projects

A remote repository already exists when

- A coauthor invites you to a team repository
- You want your own linked copy of public work
- A lab or instructor provides a reusable template
- You want to contribute a change to another repository

The starting action depends on ownership and purpose

Where we left off in Lecture 5.2

The `arec_570_demo` project retrieved land-use, streamflow, weather, and Census data through APIs

- R scripts recorded the requests
- Raw downloads preserved the responses
- A retrieval log recorded source, date, and parameters
- `docs/data-plan.md` connected the sources to a research question

Today we organize that work and begin recording its history

Follow-along repository

`arec_570_demo` develops in stages across Unit 5

Tag	Lecture	Addition
`5.2-retrieve`	API collection	Retrieval script
`5.3-structure`	Structure and Git	Folders, `run_all.R`, and `.gitignore`
`5.4-collaborate`	Collaboration	Issues, branches, and pull requests
`5.5-report`	Reporting	Codebook, sample, and data section

Completed before class

Lecture 5.2 assigned Parts One and Two of `docs/setup-rstudio.md`

- Install Git and restart RStudio
- Create a GitHub account
- Configure your Git name and email
- Create and store a GitHub personal access token
- Request, activate, and store a Census API key
- Run `usethis::git_sitrep()` without errors
- Confirm `nzchar(Sys.getenv("CENSUS_API_KEY"))` returns `TRUE`

Bring the laptop used to complete these steps

Vocabulary for shared repositories

Term	Meaning
Clone	A local copy of a remote repository and its history
Fork	A linked GitHub copy under your account
Template	An independent repository created from a standard set of files
Upstream	A conventional name for the original repository behind a fork

Branches, merges, and pull requests receive fuller treatment in Lecture 5.4

Choose how your local project begins

Situation	Starting action
You have access to a team repository	**Clone** the team repository
You want a linked copy of public or course work	**Fork**, then **clone** your fork
You want an independent project with a standard structure	**Use template**, then **clone** the new repository
You already have your own local project	**Initialize Git**, then create a GitHub remote

Cloning is the step that creates the working project on your computer

Clone a team repository

Use **clone** when the repository already exists and you have permission to work in it

In RStudio

1. Copy the repository's HTTPS URL from **Code**
2. Select **File > New Project > Version Control > Git**
3. Paste the URL and choose a parent folder
4. Select **Create Project**

Cloning copies files and history and configures the remote as `origin`

Fork public or course work

Use **fork** when you want a linked GitHub copy under your account

A fork supports

- Receiving later updates from the original repository
- Developing changes in a repository you control
- Proposing changes through a pull request

After creating the fork on GitHub, clone your fork to RStudio

Use a template for an independent project

Use **template** when a repository provides a reusable starting structure

The new repository receives

- The template's current files and folders
- A new Git history
- The repository name and visibility you choose

A personal website template is one example. Your website becomes an independent project with its own history

Which action fits the project?

A. Your advisor adds you to a private team repository

B. You want your own version of the course demo and need later course updates

C. Your department offers a personal website template

D. You have an existing dissertation project on your computer

Follow along: fork the demo

Open github.com/laurenchenarides/arec_570_demo while signed in

1. Select **Fork**
2. Clear **Copy the main branch only** so the lecture tags are included
3. Select **Create fork**
4. Confirm the address is `github.com/your-username/arec_570_demo`

Your fork remains linked to the course repository for later synchronization

Follow along: clone your fork

1. On your fork, select **Code** and copy the HTTPS URL
2. In RStudio select **File > New Project > Version Control > Git**
3. Paste the URL and choose a parent folder such as `GitProjects`
4. Select **Create Project**
5. Confirm the project name appears in the upper-right corner
6. Confirm a **Git** tab appears beside Environment and History

The local repository's `origin` points to your fork

Origin and upstream identify remotes

Name	Repository in this exercise
`origin`	Your fork on GitHub
`upstream`	The original course repository

GitHub's **Sync fork** brings course updates into your fork

After syncing the fork, **Pull** brings those commits to the local project

Inspect the change from Lecture 5.2 to 5.3

Open the comparison between `5.2-retrieve` and `5.3-structure`

Look for

- The division of one retrieval script into numbered scripts
- The addition of `run_all.R`
- The separation of raw data, derived data, and outputs
- The addition of `.gitignore`
- The retrieval log and documentation

Which change most improves another researcher's ability to reproduce the project?

Record one research decision

Create `docs/class-notes.md` and add one sentence explaining a geographic or temporal choice in the demo project

1. Inspect the changed line with **Diff**
2. Stage `docs/class-notes.md`
3. Commit with `Document geographic assignment rule`
4. Push the commit
5. Open your fork on GitHub and locate the new commit

The new file belongs to your fork, so later course updates can remain separate

Pull and push synchronize commits

Pull brings remote commits into the local repository

Push sends local commits to the remote repository

When GitHub reports newer remote commits

1. Pull the remote commits
2. Review any conflicts Git reports
3. Complete the merge if required
4. Push the combined history

Lecture 5.4 develops branches, conflicts, and pull requests

Common problems and first checks

Problem	First check
The Git tab is missing	Confirm the `.Rproj` inside a Git repository is open
Git rejects a commit	Check the configured name and email with `usethis::git_sitrep()`
GitHub rejects authentication	Check whether the personal access token is present and current
A push is rejected	Pull and inspect the incoming commits
A file never appears in the Git pane	Check the patterns in `.gitignore`

Record the exact error message when asking for help

Apply the workflow to your research project

For one current project, identify

1. The project root
2. The command or script that reproduces the main result
3. The files Git should track
4. The files `.gitignore` should exclude
5. The people who need repository access
6. The next research decision that deserves a commit

Share one unresolved choice with the class

Return to Gentzkow and Shapiro

How do the reading's recommendations appear in the demo repository?

- Automation through `run_all.R`
- Separate directories for code, raw data, derived data, and outputs
- Relative paths from the project root
- Version history for code and documentation
- README files that describe execution and file roles

Which recommendation requires the largest change in your current workflow?

Before you leave

Confirm that your fork contains one new commit from today's exercise

Be prepared to show

- The line you changed
- The commit message
- The same commit in RStudio and on GitHub
- One file your research repository should exclude

Summary

Project structure records where inputs, code, and outputs belong

Git records local changes as commits

GitHub hosts a remote repository for sharing and collaboration

RStudio provides an interface for editing files and using Git

Clone, fork, and template describe different ways to begin from an existing repository

Pull, edit, inspect, stage, commit, and push form the everyday workflow

References

Bryan, Jennifer. 2017. “Excuse Me, Do You Have a Moment to Talk about Version Control?” *PeerJ Preprints* 5: e3159v2. https://doi.org/10.7287/peerj.preprints.3159v2.

Gentzkow, Matthew, and Jesse M. Shapiro. 2014. *Code and Data for the Social Sciences: A Practitioner’s Guide*. University of Chicago mimeo. https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf.

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